AKADEMIKA

BCL-FSK

FSK MODULATION / DEMODULATION

KIT

EXPERIMENTAL MANUAL

…. A MESSAGE FROM

AKADEMIKA

Today's system designers are faced with tomorrow's problems. DIGITAL COMMUNICATION is one of the important subject need to teach while learning electronics.
It is our vision to provide you with the product you need for training ensuring lasting reliability & quality.

OUR MOTTO;

- Light years ahead – refers to leadership.

As leaders in our industry in India, we are totally committed to servicing as the standard against which all are measured in the areas of

• Design • Quality • Value • Delivery • Support.

We are truly light years ahead of our competition in this area. That means that you our valued customers are guaranteed satisfaction.

As you will move through this manual you will quickly discover that we have complete, truly innovative & superior training products. We are so committed to quality that we back our products with a complete comprehensive warranty.

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SAFETY RULES

Carefully follow the instructions contained in this handbook as they give important points on safety during the installation, use and maintenance. Keep this handbook always with you for any further help.

After the unpacking, set all accessories in order so that they are not lost and check the kit integrity. In particular, check that the kit is integral and shows no visible damage.

Before connecting the power supply to the kit, be sure that the jumpers and patch chords are connected correctly as per experiment.

In BCL-FSK kit, signals are to be monitored with an oscilloscope or with help of indicators as explained in the manual.

This kit must be employed only for the use for which it has been conceived, i.e. as educational equipment and must be used under the direct survey of expert personnel. Any other use is improper and so dangerous. The manufacturer cannot be considered responsible for eventual damages due to improper, wrong or unreasonable uses.

In case of any fault/bad operation, turn off the power supply and do not tamper the kit. In case servicing is required, contact the service center for technical assistance.

The kits are liable to malfunction/under-perform if they are not operated under standard environmental conditions of temperature and humidity**.**

WARRANTY

This kit is warranted against defects in workmanship and materials. Any failure due to defect in either workmanship or material should occur under normal use within a year from the original date of purchase, such failure will be corrected free of charge to the purchaser by repair or replacement of defective part or parts. When the failure is result of user's neglect, natural disaster or accident, we charge for repairs regardless of the warranty period. The warranty does not cover perishable items like connecting chords, Crystals, etc. and other imported items.

This warranty is subject to the following conditions and limitations. The warranty is void and inapplicable if the defective product is not brought or sent to our authorized service center or sales outlet within the warranty period. Defective product is, on Akademika Lab Solutions sole judgment. The defective product will be replaced with new one or repaired without charge or with charge.

In the warranty period if the service is needed, purchaser should get in touch with the service center or sales outlet. The purchaser should return the product to the service center or sales outlet at his or her sole expense. The loss and damage in transit will be outside the preview of this warranty. A returned product must be accompanied by a written description of the defects. Type and Model No. of kits has to be mentioned specifically. We return the product to the purchaser at our expense. In case that warranty does not cover the product on Akademika Lab Solutions judgment, we repair the product after obtaining prior permission from purchaser who receives an estimate statement about repairing charges. In such case, Akademika Lab Solutions bares the transporting expenses required to send back all the repaired products for the moment, and then repairs and transporting expenses will be charged against the purchaser by the statement of accounts.

When the authorized sales agents sell our products, they must notify the purchaser of the warranty contents, but have no rights to stretch the meaning of original warranty contents or offer additional warranty. Akademika Lab Solutions does not provide any other promise or suggestive warranty and hold no liability for the damage caused by negligence, abnormal use or natural disaster. Akademika Lab Solutions is not responsible for the damages even if it is notified of above dangers in advance as well.

For more special service or overall repairs, maintenance and up gradation of products, be sure to contact our service center or sales outlet.

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TECHNICAL SPECIFICATION

BCL-FSK: FREQUENCY SHIFT KEYING MODULATIOPN/DEMODULATION

DATA SIMULATOR : Onboard 8-bit variable NRZ-L pattern.

CRYSTAL OSCILLATOR : 2 MHz

DATA CLOCK : 250 KHz.

ON BOARD CRRIER SINE WAVES : 1 MHz (0°) and 500KHz, 4VPP

CARRIER MODULATION : FSK.

CARRIER DEMODULATION : FSK

INTERMEDIATE SIGNAL : Provision for observing intermediate signals during demodulation.

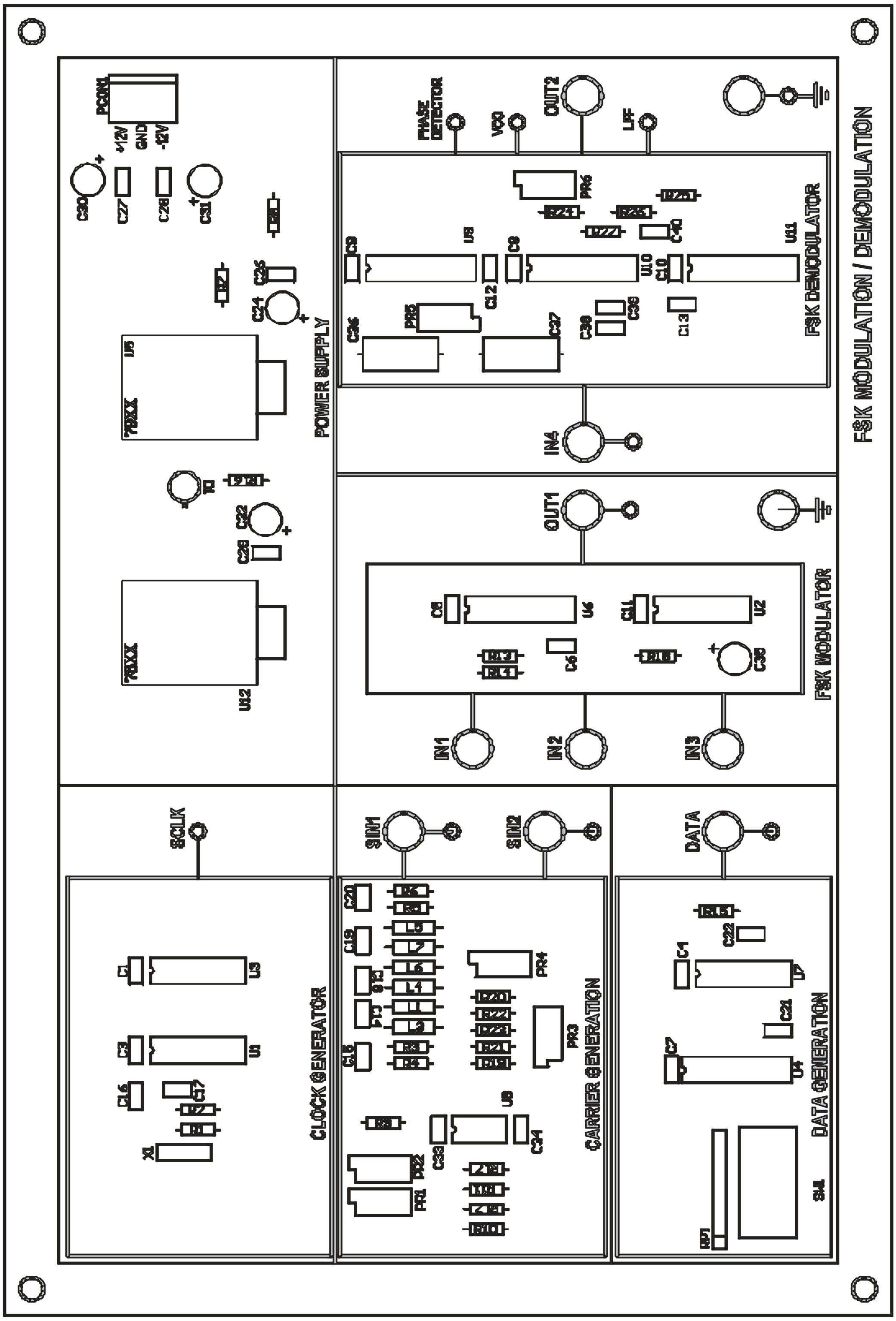
INTER CONNECTION : 2 mm banana socket.

POWER SUPPLY : +12V, -12V, GND.

ACCESSORIES

	BCL-FSK	Quantity
1	Short Link Patch Cords male to male	4
2	12V Power Adapter	1
3	BCL-FSK EXPT Manual	1

FUNCTIONAL BLOCK



INTRODUCTION

"BCL-FSK" initiates the user to the various Carrier Modulation Techniques, normally adopted in practice

"BCL-FSK": FREQUENCY SHIFT KEYING MODULATOR / DEMODULATOR KIT.

FEATURES:

1) The trainer comes with an onboard data simulator, which generates the NRZ-L pattern depending on positions of 8-bit switch and the reference clock (256 KHz), which enables the trainers to work in a stand-alone mode.

2) Carrier modulation options are available on the kit. Frequency Shift Keying (FSK)

3) Onboard carrier generation circuit generates sine waves synchronized to the transmitted data. The three carrier sine waves generated onboard are of frequencies.
    a) 1.024KHz (0°)
    b) 512KHz (0°)

4) Interconnection facilities:
   Sockets and patch chords are provided for on board connections.

5) Test Points:
   All relevant test points are brought-out for observations. Observations are carried out on an oscilloscope.

6) Power Supply:
   -12V, +12V, GND.

PRINCIPLES OF DIGITAL MODULATION AND DEMODULATION TECHNIQUES:

Digital Communication is the technique by which the whole information is represented in terms of binary digits, i.e., `ones' and `zeros'. These digits are represented by discrete voltage levels and the clock frequency of a digital communication scheme is generally low. For long distance transmission, the data is made to modulate a continuous wave (sine wave) carrier. These techniques are called Carrier Modulation Techniques. The various types of Carrier Modulation Techniques normally adopted in practice fall under three broad categories.

Frequency Shift Keying (FSK),

FREQUENCY SHIFT KEYING (FSK):

In this type of modulation, the modulated output shifts between two frequencies for all `one' to `zero' transitions.

Let the carrier frequencies be represented by w1 and w2, and then we have:

M (t)	=	A (t) Cos w1t	if data is `One'
		A (t) Cos w2t	if data is `Zero'.
Where A (t)	=	Time varying amplitude of the sinewave.	
M (t)	=	Modulated carrier.	

FSK Demodulator employs PLL logic for the recovery of data.

EXPERIMENT
NO. 1

EXPERIMENT NO: 1

Aim:

To study the Frequency Shift Keying Modulation and Demodulation.

Equipment:

FSK board, power supply, patch chord, Dual channel oscilloscope/DSO.

Theory:

Frequency shift keying is relatively simple, low performance type of digital Modulation. Binary FSK modulation is constant amplitude angle modulation similar to conventional FM expect that the modulating signal is a binary signal that varies between two distinct voltage levels rather than the continuously changing analog signal.

In FSK the carrier frequency (center) is chosen such that it falls halfway between the mark and space frequency. Logic ‘1’ input shifts the VCO output to the mark frequency, and logic ‘0’ to the space frequency. The VCO output shifts or deviates back and forth between the mark and space frequencies.

At the demodulation the FSK input signal is simultaneously applied to the inputs of both band pass filters through the power splitter. The respective filter passes only mark and space frequencies on to its respective envelope detector. The envelope detectors, in turn indicate the total power in each pass band and comparator responds to the larger of the two powers. This type of FSK detection is referred to as non coherent detection; there is no frequency involved in the demodulation process that is synchronized either in phase, frequency, or both with the incoming signal FSK signal.

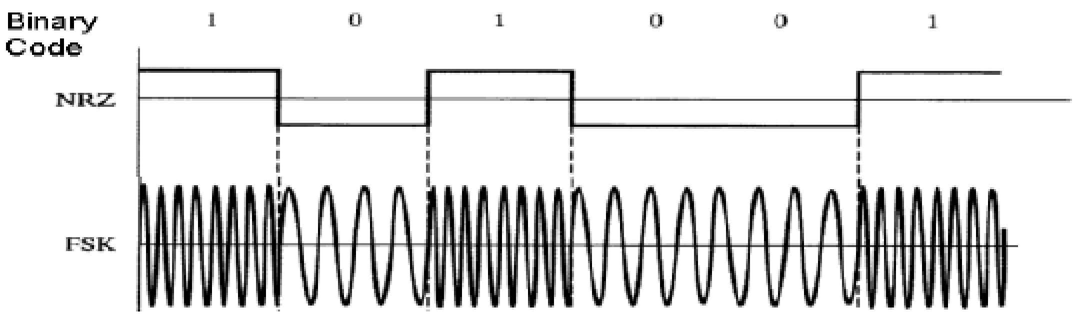


Fig 2. FSK waveforms

Applications of FSK:

In 1910, Reginald Fessenden invented a two-tone method of transmitting Morse code. Dots and dashes were different tones of equal length. The intent was to minimize transmission time.

Some early CW transmitters employed an arc converter that could not be conveniently keyed. Instead of turning the arc on and off, the key slightly changed the transmitter frequency in a technique known as the *compensation-wave method*. The compensation-wave was not used at the receiver. The method consumed a lot of bandwidth and caused interference, so it was discouraged by 1921.

Most early telephone-line modems used audio frequency-shift keying (AFSK) to send and receive data at rates up to about 1200 bits per second. The common Bell 103 and Bell 202 modems used this technique. Even today, North American caller ID uses 1200 baud AFSK in the form of the Bell 202 standard. Some early microcomputers used a specific form of AFSK modulation, the Kansas City standard, to store data on audio cassettes. AFSK is still widely used in amateur radio, as it allows data transmission through unmodified voice band equipment. Radio control gear uses FSK, but calls it FM and PPM instead.

AFSK is also used in the United States' Emergency Alert System to transmit warning information. It is used at higher bitrates for Weather copy used on Weatheradio by NOAA in the U.S.

The CHU shortwave radio station in Ottawa, Canada broadcasts an exclusive digital time signal encoded using AFSK modulation. FSK is commonly used in Caller ID and remote metering applications

Procedure:

1. Connect power supply to kit with proper polarity and switch it ON.
2. Refer to KIT and carry out connections as mentioned below.
3. Observe carrier frequency at SIN1(1MHz) and SIN2 (500KHz) posts and note down the same.
4. Set data pattern as 10111001 using switch SW1 at DATA post.
5. Connect SIN1 post to IN2 post and SIN2 post to IN1 using patch chord.
6. Connect DATA post to IN3 post.
7. Observe FSK modulated signal at OUT1 post. Compare it with DATA. It is similar to fig 2 shown above.
8. To demodulate this signal connect OUT1 post to IN4 post.
9. FSK demodulator uses PLL detector technique to demodulate the FSK signal.

Observe recovered data pattern at OUT2 post. Compare it with DATA